

Cost Arbitrage in Multi-Cloud AI

Optimizing spend by choosing the right provider for the right job

Strategic cloud selection can dramatically reduce AI infrastructure costs while maintaining performance. This presentation explores how to leverage price differences across cloud providers to optimize your AI budget.



What Is Cost Arbitrage?



Strategic Multi-Cloud

Deliberately using multiple cloud providers to take advantage of pricing differences and specialized offerings



Balance Factors

Making decisions based on the optimal mix of **price**, **performance**, and **availability** for each workload



Critical for AI

Especially valuable for **GPU-intensive AI workloads** where infrastructure costs represent a significant budget item

Why It Matters for AI



AI infrastructure costs can vary dramatically between providers

High-Cost Components

GPUs, high-performance storage, and specialized networking represent significant budget line items

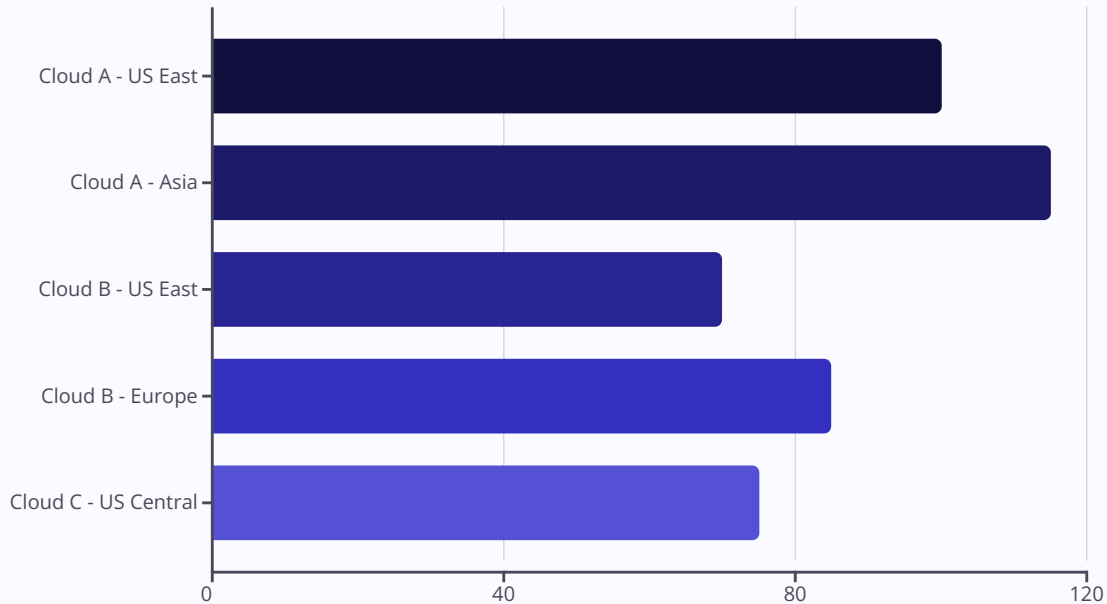
Pricing Variability

Substantial differences in pricing across providers, regions, and commitment terms

Discount Opportunities

Leverage [spot instances](#), [preemptible VMs](#), and reserved capacity discounts

Example: GPU Pricing Variability



Same GPU hardware can cost 30-40% less depending on which cloud provider and region you choose

Arbitrage Strategy: Train models in the **lowest-cost region/provider**, then deploy for inference closer to users

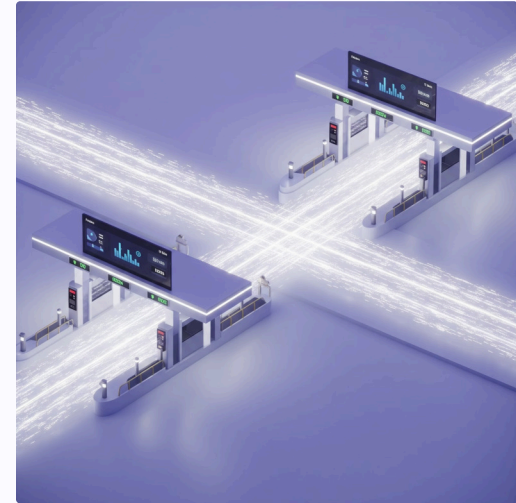
Storage & Network Arbitrage

Storage Arbitrage

Object Storage: GCS offers better pricing for bulk archival, while S3 has better ecosystem tool integration

Block Storage: Performance/price ratios vary by 15-25% across providers for similar IOPS levels

Network Costs



Network egress is often the hidden cost driver that can erase arbitrage savings

Minimize cross-cloud data movement to maximize arbitrage benefits

Multi-Cloud Arbitrage Strategies



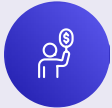
Bursting

Use secondary cloud for additional capacity during demand spikes when primary cloud costs rise



Workload Splitting

Training on cloud with cheapest GPUs, inference on cloud closest to users



Spot/Preemptible VMs

Shift fault-tolerant workloads to whichever provider offers lowest spot pricing



SaaS Arbitrage

Choose different managed AI services based on pricing efficiency for specific model types

The most effective arbitrage strategies combine multiple approaches and automate decision-making based on real-time pricing.

Tooling for Arbitrage



Infrastructure as Code

Terraform and Pulumi enable portable deployments that can be shifted between providers



Kubernetes Federation

Simplifies moving containerized AI workloads between clusters on different clouds



Cost Monitoring

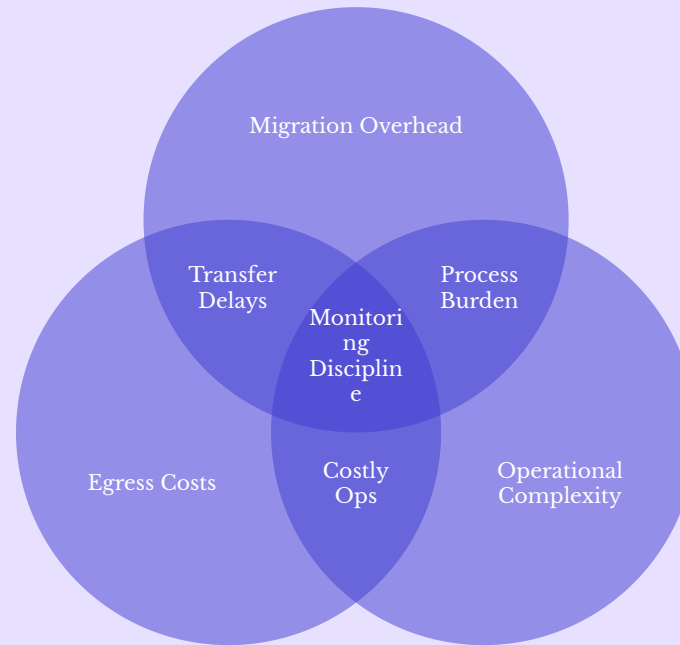
CloudHealth, Kubecost, and custom FinOps dashboards track arbitrage opportunities



Automated Bidding

Custom autoscalers that place workloads on cheapest available resources

Risks & Challenges



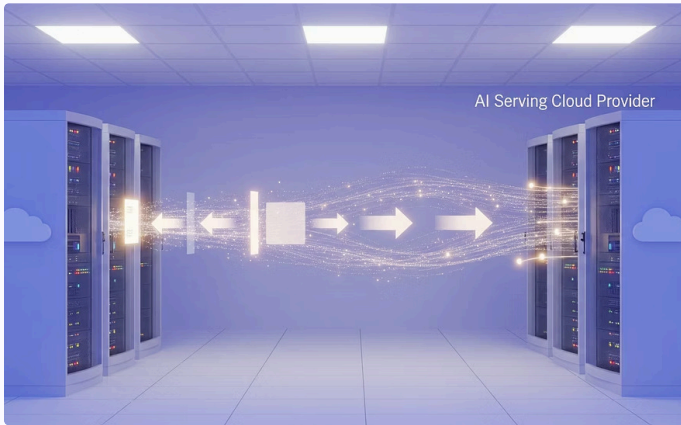
Egress Costs: Data transfer between clouds can quickly erase GPU savings – 1TB of egress costs \$50-120

Operational Complexity: Managing multiple dashboards, SLAs, and APIs increases team cognitive load

Migration Overhead: Moving workloads takes engineering time that must be factored into TCO

Monitoring Discipline: Requires rigorous cost tracking and automation to avoid "cloud sprawl"

Real-World Examples



LLM Startup

Trains models on [spot GPUs in GCP](#) (saving 60-70%), serves inference on [AWS](#) to leverage enterprise connectivity options



Gaming AI Company

Dynamically shifts NPC behavior training between [Azure and AWS GPUs](#) based on hourly price fluctuations, saving 22% annually



Enterprise Negotiation

Fortune 500 company demonstrated multi-cloud capability to negotiate an additional [15% discount](#) on their primary cloud provider

Best Practices

1 Calculate Total Cost of Ownership

Include egress, operations time, and monitoring overhead – not just raw VM pricing

2 Automate Placement Decisions

Combine FinOps data with orchestration to make real-time arbitrage decisions

1 Develop Egress-Aware Data Strategy

Design architecture to minimize cross-cloud data movement

2 Leverage for Vendor Negotiations

Use multi-cloud capability as bargaining leverage for better discounts

📌 **Pro Tip:** Document your arbitrage strategies as "playbooks" that can be executed when specific pricing thresholds are crossed.



Key Takeaways

Powerful Cost Lever

Multi-cloud arbitrage provides significant cost control for AI/ML operations at scale

Focus Areas

Target [GPUs, storage, network egress, and managed AI services](#) for highest ROI

Execution Requirements

Success depends on automation and a strong FinOps culture with real-time visibility

Financial Impact

When implemented correctly, can save [millions annually](#) for organizations with large AI workloads